

Lab 2 (Part 1): Forensic Bias Audit

Deliverable 2: Impossibility Brief

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Forensic Bias Audit of a COMPAS-like Recidivism Classifier: An Impossibility Brief

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1. Empirical Audit Findings

This audit is not a historical exercise. In June 2026, California's Department of Technology confirmed that COMPAS is among six systems the state now classifies as high-risk automated decision-making, used by the California Department of Corrections and Rehabilitation to assign recidivism scores for at least a decade (Johnson, 2026). The corrections department's own prior submissions had not characterized this use as high-risk, and Senate Bill 1248, which would have barred state employees from using an automated system as the sole basis for a consequential decision, was killed in this year's budget process (Johnson, 2026). The system audited here is, in a direct sense, the system currently assigning risk scores to Californians awaiting judgment.

This audit examined a COMPAS-like recidivism classifier using Google's What-If Tool (Google PAIR, n.d.), testing individual reliability near the decision boundary and group level fairness across race. Three near threshold data points (342, 464, and 894) were selected with baseline inference scores between 0.498 and 0.506, each within 0.008 of the 0.5 threshold. Increasing `priors_count` from 1 to 2 raised the inference score in all three trials (+0.098, +0.110, +0.109), with no non-monotonic behavior observed. Datapoint 342 crossed the threshold entirely, moving from Low Risk (0.498) to High Risk (0.595), demonstrating that classification outcomes for borderline defendants can hinge on a single incremental feature change rather than a stable overall risk profile. A nearest counterfactual comparison (L1 distance) on Datapoint 464 in its post edit state (`priors_count` = 2, score 0.616) returned Datapoint 2047 (Low Risk, score

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0.453) as the closest oppositely labeled record. Of nine recorded features, only `priors_count` differed (2 versus 0); age, race, sex, and juvenile-history fields were identical, indicating that for this instance the local decision boundary was governed by criminal-history features rather than a demographic proxy.

Group level auditing sliced performance by race. At baseline 0.5 threshold for both groups, African-American defendants ($n = 1,904$) faced a False Positive Rate (FPR) of 24.5% and False Negative Rate (FNR) of 11.4%, against a base rate of 53.5%. Caucasian defendants ($n = 1,111$) faced an FPR of 9.6% and FNR of 23.7%, against a base rate of 42.8%. The combined disparity term, $D_{\text{fair}} = |\text{FPR}_{\text{AA}} - \text{FPR}_{\text{C}}| + |\text{FNR}_{\text{AA}} - \text{FNR}_{\text{C}}|$, was 27.2 points at baseline. Applying an Equal Opportunity constraint shifted classification thresholds to 0.59 African-American and 0.39 Caucasian, narrowing D_{fair} to 4.1 points, an 85% reduction, at a weighted accuracy cost of 0.63 percentage points (65.1% to 64.4%). This "ethical tax" reflects a small predictive cost that substitutes one error pattern, over flagging African-American defendants as High Risk, for another, under flagging Caucasian reoffenders, without eliminating error altogether.

2. Theoretical Analysis: Kleinberg's Impossibility Theorem

Calibration means predicted risk scores correspond to observed outcome frequencies within each group, among defendants scored 70% risk, roughly 70% actually reoffend, independently by race. Equal Opportunity requires equal True Positive Rates across groups, so an actual reoffender is equally likely to be correctly flagged High Risk regardless of race. Kleinberg et al. (2016) proved that when base rates differ across groups, a classifier cannot satisfy calibration and an equalized error constraint such as Equal Opportunity simultaneously, except in the trivial case of a perfect predictor. This audit's own data instantiates the theorem

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directly: base rates were 53.5% for African-American defendants versus 42.8% for Caucasian defendants, a 10.7-point gap. Because these rates differ, and the model is far from perfect (AUC 0.71), satisfying both properties at once is mathematically foreclosed, not merely difficult to engineer.

The baseline audit reflects a roughly calibration consistent model, since both groups were evaluated at the same 0.5 threshold, paired with sharply divergent error rates. This pattern is not an artifact of the WIT demo or its particular slice of the dataset. A 2026 replication of the original ProPublica COMPAS data using modern classifiers, including Logistic Regression, Support Vector Machines, and XGBoost, found the same asymmetry, elevated FPR for African-American defendants and elevated FNR for Caucasian defendants, across every classifier and every debiasing technique tested, including explicit removal of the race feature (Kienzle et al., 2026). Adding or removing race made no statistically detectable difference to outcomes, indicating the disparity is carried by correlated proxy features rather than the race field itself: independent, current confirmation that the pattern audited here is structural to the data, not a quirk of one modeling choice. Equal Opportunity narrowed the FNR gap from 12.3 to 3.9 points, but only by separating group thresholds, meaning an identical risk score is no longer treated identically across race, which is itself a loss of calibration. This is not a tuning failure; it is the structural trade off the theorem predicts. Any correction toward equal error rates under unequal base rates necessarily costs calibration, and any preservation of calibration necessarily leaves error rates unequal. Choosing between these representations is a governance decision, not a modeling one.

This audit's disparity is properly understood as both a statistical constraint and a symptom of structural inequity. The statistical constraint is Kleinberg's result itself: given the observed

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base rate gap, no threshold choice can equalize both calibration and error rates. Structural inequity concerns why the base rates differ in the first place, a 53.5% versus 42.8% recidivism base rate is shaped by unequal policing intensity and socioeconomic exposure to the justice system that predate any modeling decision. Tuning λ or applying Equal Opportunity can redistribute the statistical consequence, but it cannot repair the structural cause upstream in the data-generating process.

3. Loss Function Formalization and Mathematical Justification of λ

The objective $\min_{\theta} J(\theta) + \lambda * D_{\text{fair}}(P_{\theta})$ formalizes this governance decision, where $J(\theta) = 1 - \text{Accuracy}$ and $D_{\text{fair}}(P_{\theta}) = |\text{FPR}_{\text{AA}} - \text{FPR}_{\text{C}}| + |\text{FNR}_{\text{AA}} - \text{FNR}_{\text{C}}|$. At $\lambda = 0$, the model optimizes purely for accuracy with no fairness penalty, producing the baseline result, 65.1% weighted accuracy against a D_{fair} of 27.2. As λ approaches infinity, the objective prioritizes eliminating D_{fair} regardless of accuracy cost, and at the extreme could force degenerate solutions that ignore predictive signal entirely.

The empirical Equal Opportunity result, D_{fair} falling from 27.2 to 4.1 for a 0.63-point accuracy cost, behaves as though λ were set to a moderate, well-chosen value: large enough to compress disparity substantially, small enough to preserve nearly all predictive power. This audit defends that implicit weighting on normative grounds rather than mathematical ones: an 85% reduction in cross racial error disparity is worth well under a point of accuracy because the harm being reduced, wrongful High Risk labeling, is a liberty harm falling disproportionately and involuntarily on one group, while the accuracy cost is small and diffusely distributed across the population. A jurisdiction with a stronger public safety mandate could reasonably set λ lower and accept more disparity; that is a legitimate but different policy choice, not a technical error.

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Governance Parameter / Metric	Baseline Model ($\lambda=0$)	Equal Opportunity Model ($\lambda>0$)	NIST AI RMF Target / Trigger
Primary Objective	Maximize Predictive Accuracy	Equalize Group TPR / FNR	Audit Boundary & Group Disparities
Weighted Accuracy	65.06%	64.43%	Ethical Tax limit: $< 1.0\%$
Disparity Metric (D_{fair})	27.2 points	4.1 points	Re-audit trigger if $D_{\text{fair}} > 10.0$
African-American FPR / FNR	24.5% / 11.4%	18.0% / 19.0%	Flag if FPR disparity $> 15.0\%$
Caucasian FPR / FNR	9.6% / 23.7%	18.2% / 15.1%	Flag if FNR disparity $> 12.0\%$
Human Oversight Window	None (unmanaged)	None (threshold shifted only)	Mandatory review for $0.45 \leq \text{Score} \leq 0.55$

Table 1. Summary of loss-function trade-offs (Section 3) and NIST AI RMF governance triggers (Section 4) across the baseline and Equal Opportunity configurations.

4. Governance Alignment (NIST AI RMF Playbook & PTRa Framework)

Three suggested actions from the NIST AI RMF Playbook (National Institute of Standards and Technology [NIST], 2023) address the findings above. GOVERN 1.2 (Trustworthy Characteristics) includes "detail model testing and validation processes" as a suggested action (NIST, 2023). Requiring documented boundary-sensitivity and counterfactual probes as a pre-deployment validation gate would catch boundary fragility and counterfactual sensitivity before real cases are affected, and requiring sliced FPR and FNR by race as a pass or fail criterion would prevent a model exhibiting a 14.9 point FPR gap from reaching deployment undocumented. This requires sign-off from a validation function independent of development, with retained test records; documentation should name the base-rate gap as likely reflecting structural inequities in policing, not merely a statistical artifact. A reasonable monitoring trigger is re-validation if D_{fair} exceeds 10 points or the base rate gap shifts more than 5 points from the training distribution.

A second GOVERN 1.2 action, "establish the frequency of and detail for monitoring, auditing and review processes" (NIST, 2023), addresses drift over time. Periodic resampling of

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near threshold cases and scheduled recomputation of sliced metrics would catch boundary instability or disparity that reemerges after a threshold based correction decays. This requires a named auditing function, a defined cadence such as quarterly review, and an escalation path to legal and compliance. If disparity persists even as base rates converge, that increasingly implicates the model or deployment context rather than the statistical constraint alone. A reasonable trigger is re-audit if FPR disparity exceeds 15 points or FNR disparity exceeds 12 points at any checkpoint.

GOVERN 3.2 (Human-AI Teaming) includes "establish policies and procedures regarding AI actor roles, and responsibilities for human oversight of deployed systems" (NIST, 2023). Mandatory human review for scores within a defined margin of threshold, such as 0.45 to 0.55, directly targets the boundary fragility this audit identified, ensuring the score alone is not dispositive for the population most affected by single-feature instability. Reviewers with visibility into known FPR and FNR asymmetries can apply calibrated judgment, and reviewers trained to request a nearest counterfactual comparison on borderline cases can escalate any where race appears among the differing features. This requires documented reviewer qualifications, a logged record of every override, and periodic review of override patterns for their own racial skew. Human oversight cannot resolve the statistical constraint itself, since the calibration and Equal Opportunity trade off is mathematically fixed given the base rates, but it can catch cases where structural inequities in the underlying data are being mechanically reproduced by an unexamined score. A reasonable trigger is mandatory escalation for any score within 0.05 of the active threshold, and policy review if override rates differ by more than 10 points across race groups over a rolling 90-day window.

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A useful comparison case for this recommendation is the federal Pretrial Risk Assessment (PTRA), a different structured tool applied to federal defendants. Modeling a policy that presumptively releases PTRA classified low risk defendants, rather than deferring entirely to unstructured judicial discretion, is estimated to reduce detention by 34.2% while narrowing, not widening, the racial gap, Black defendants would see a larger detention reduction (39.0%) than White defendants (27.3%) (Skeem et al., 2025). The authors caution that the tool should strongly guide, not replace, judicial discretion. The difference between that projected outcome and COMPAS's current unmanaged deployment in California is not the algorithm; both tools face the same base rate constraint described in Section 2. It is governance, a specified release policy paired with retained human discretion, exactly what GOVERN 3.2 recommends, is what turns a structured score into a disparity reducing tool rather than a disparity-reproducing one.

5. Conclusion

The empirical audit and the theoretical synthesis point to the same conclusion: the disparities observed in this classifier are not implementation bugs correctable by better tuning, but structural consequences of applying a single threshold to groups with different base rates, a pattern independently confirmed in fresh 2026 replication of the underlying data. Equal Opportunity purchased a large fairness gain, an 85% reduction in combined error-rate disparity, for a small accuracy cost of 0.63 points, which is a defensible governance choice. But it is a choice, not a neutral technical fix, it redistributes who is wrongly detained and who is wrongly released, and leaves the underlying base-rate gap untouched. As the PTRA comparison shows, the same statistical constraint can produce a disparity-reducing outcome or a disparity-reproducing one depending entirely on governance, documented predeployment validation, a recurring audit cadence, and accountable human oversight for the narrow band of

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cases this audit found most fragile. With COMPAS now confirmed in active use across California's corrections system, absent all three, a mathematically defensible λ is not sufficient to make the system trustworthy in practice.

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